

Noorinder

1. File Not Found

- **Error:** The program crashes with a `FileNotFoundException` when the data file `piggybank.dat` doesn't exist.
- **Cause:** The program assumes the file is always present.
- **Fix:** Initialize a new `PiggyBank` object if the file is missing.

Before Code:

```
FileInputStream fileIn = new FileInputStream("../Chapter11/src/Mastery/piggybank.dat");
```

After Fix:

```
try {
    FileInputStream fileIn = new FileInputStream("../Chapter11/src/Mastery/piggybank.dat");
    ObjectInputStream readBank = new ObjectInputStream(fileIn);
    piggyBank = (PiggyBank) readBank.readObject();
    readBank.close();
} catch (FileNotFoundException e) {
    System.out.println("File not found. Starting with a new PiggyBank.");
    piggyBank = new PiggyBank();
}
```

2. Data Corruption

- **Error:** The file becomes corrupted if the program crashes while saving the `PiggyBank` data.
- **Cause:** The `ObjectOutputStream` wasn't closed properly during the save operation.
- **Fix:** Used `try-with-resources` to ensure the stream always closes.

Before Code:

```
ObjectOutputStream writeBank = new ObjectOutputStream(fileOut);
writeBank.writeObject(piggyBank);
writeBank.close();
```

After Fix:

```
|
try (ObjectOutputStream writeBank = new ObjectOutputStream(fileOut)) {
    writeBank.writeObject(piggyBank);
} catch (IOException e) {
    System.err.println("Error saving PiggyBank data: " + e.getMessage());
}
```

3. Invalid Input

- **Error:** The program crashes when the user enters non-numeric input.
- **Cause:** The program doesn't validate the input before using it.
- **Fix:** Added input validation to ensure only numbers are accepted.

Before Code:

```
choice = scanner.nextInt();|
```

After Fix:

```
try {
    System.out.print("Enter your choice: ");
    choice = Integer.parseInt(scanner.nextLine());
} catch (NumberFormatException e) {
    System.out.println("Invalid input. Please enter a number.");
    choice = -1;
}
```

4. Serialization Compatibility

- **Error:** The program fails to load older files after changes to the `PiggyBank` class.
- **Cause:** No `serialVersionUID` was set, breaking compatibility between versions.
- **Fix:** Added a `serialVersionUID` to the `PiggyBank` class.

Code Example:

```
private static final long serialVersionUID = 1L;|
```

